Microsoft Elevate

CAPSTONE PROJECT

CUSTOMER BEHAVIOUR ANALYSIS

PRESENTED BY

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PROBLEM STATEMENT:

- Businesses collect large amounts of customer transaction data but often struggle to convert this data into meaningful insights.
- Understanding customer purchasing behavior, product preferences, and spending patterns is essential for improving marketing strategies and increasing sales.
- However, without proper analysis tools and visualization techniques, organizations find it difficult to identify trends in customer data.
- Therefore, analyzing customer shopping behavior using data analytics techniques is important for helping businesses make data-driven decisions and improve customer satisfaction.

PROPOSED SOLUTION:

- The proposed system aims to address the challenge of understanding customer purchasing patterns and identifying key factors that influence shopping behavior. This involves leveraging data analytics techniques to analyze transactional data and generate meaningful insights that support business decision-making. The solution will consist of the following components:

- **Data Collection:**
  Gather customer shopping data that includes demographic details, product categories, purchase amounts, review ratings, subscription status, and shipping types.
  The dataset provides historical transaction information that helps analyze customer preferences and purchasing trends.

- **Data Preprocessing:**
  Clean and preprocess the collected data to handle missing values, remove inconsistencies, and standardize column formats.
  Perform feature engineering by creating new variables such as age groups and purchase frequency to improve analysis and extract deeper insights from the data.

- **Deployment:**
  Develop an interactive Power BI dashboard that visually presents customer behavior insights such as revenue by category, sales by age group, and subscription distribution.
  Deploy the dashboard on Power BI Desktop or Power BI Service so that stakeholders can easily access and analyze customer trends to support data-driven business decisions.

SYSTEM APPROACH:

- The "System Approach" section outlines the overall strategy and methodology used to develop and implement the customer behavior analysis system. The approach focuses on collecting customer shopping data, processing it using data analytics techniques, and presenting insights through interactive visualizations.
- **System Requirements:**
- A computer with Python installed
- Jupyter Notebook or any Python IDE
- SQL Database (PostgreSQL / MySQL / SQL Server)
- Power BI Desktop for dashboard visualization
- Basic hardware requirements such as minimum 4GB RAM and internet connectivity
- **Libraries Required to Build the System:**
- **pandas** – for data loading, cleaning, and manipulation
- **numpy** – for numerical computations
- **matplotlib / seaborn** – for data visualization during analysis
- **psycopg2 / SQL connector** – for connecting Python with SQL database
- These tools and libraries help in performing data preprocessing, exploratory analysis, SQL-based business queries, and creating an interactive dashboard for visualizing customer behavior insights.

ALGORITHM & DEPLOYMENT:

- **Algorithm Selection:**
  The project uses SQL Aggregation Algorithms (SUM, AVG, COUNT), Window Function (ROW_NUMBER), and Rule-based Customer Segmentation using CASE statements to analyze customer purchase behavior and identify trends.
- **Data Input:**
  Customer dataset containing features such as Age, Gender, Category, Item Purchased, Purchase Amount, Review Rating, Subscription Status, Shipping Type, and Previous Purchases.
- **Analysis Process:**
  Data is cleaned using Python (pandas) and stored in a SQL database where aggregation queries and segmentation rules are applied to extract insights like revenue by gender, top products, and customer segments.
- **Deployment:**
  The final insights are visualized through an interactive Power BI dashboard, allowing users to explore sales trends, revenue distribution, and customer behaviour using filters and charts.

RESULT:

Customer Behavior Dashboard
Subscription Status
No
Yes
Gender
Female
Male
Category
Accessories
Clothing
Footwear
Outerwear
Shipping Type
2-Day Shipping
Express
Free Shipping
Next Day Air
Standard
Store Pickup
3.9K
Nummber of customers
3.75
Average Review Rating
₹ 59.76
Average Purchase Amount
% Of Customers By Subscription Status
Yes 27%
No 73%
subscription_st...
No
Yes
Revenue By Category
0.1M
0.0M
Clothing
Accesso...
Footwear
Outerwear
Sales By Category
2K
1K
0K
Clothing
Accesso...
Footwear
Outerwear
Revenue By Age Group
Young Adult
Middle-Aged
Adult
Senior
0K
20K
40K
60K
Sales By Age Group
Young Adult
Middle-Aged
Senior
Adult
0
500
1,000

CONCLUSION:

- The project successfully analyzes customer transaction data to identify patterns such as product category performance, customer segments, and revenue distribution across different age groups.
- The use of Python for data preprocessing, SQL for analytical queries, and Power BI for visualization helps convert raw data into meaningful business insights. Some challenges encountered during implementation included handling missing values, cleaning inconsistent data, and preparing the dataset for analysis.
- Overall, the project demonstrates how data analytics and interactive dashboards can help businesses better understand customer behavior, improve marketing strategies, and support data-driven decision making.

FUTURE SCOPE:

- The project can be improved by incorporating additional data sources such as real-time transaction data, customer browsing behavior, and seasonal sales trends to gain deeper insights into customer purchasing patterns.
- The analysis can be further optimized by applying advanced machine learning algorithms to predict customer purchase behavior, recommend products, and identify potential high-value customers. Expanding the system to analyze data from multiple stores, regions, or online platforms can provide a more comprehensive view of customer behavior.
- Future improvements may also include integrating cloud-based analytics platforms, real-time dashboards, and advanced technologies such as artificial intelligence and predictive analytics to support more efficient and intelligent business decision-making.

REFERENCES:

•Customer Shopping Behavior Dataset – Kaggle Dataset Repository
•Python Documentation – https://docs.python.org
•Pandas Library Documentation – https://pandas.pydata.org/docs
•PostgreSQL Documentation – https://www.postgresql.org/docs
•Microsoft Power BI Documentation – https://learn.microsoft.com/power-bi
•Customer Shopping Behavior Analysis Project Report and SQL Queries

GITHUB LINK:https://github.com/varshith2323/Customer-Behaviour-Analysis

Thank You